Summarization: A GP-based Robust Motion Planning Framework for Agile Autonomous Robot Navigation and Recovery in Unknown Environments

Summarized by Xiaonan (Nice) Wang *
Summarization†Generated based on Paper from Mohammad, Higgins and Bezzo (2024)
Topic: Motion Planning, Safety-Guaranteed and Collision-Free Navigation, Recovery Mechanism

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Abstract

This document summarizes the core contributions and methodology of the paper "A GP-based Robust Motion Planning Framework for Agile Autonomous Robot Navigation and Recovery in Unknown Environments, Mohammad et al. [2024]", focusing on its' main ideas and the core blocks.

1 Overview: Core Questions and Answers

(1) What is the problem?

- Robust Agile Robot Navigation in **Cluttered, Unknown** Environments
- **Safety Monitoring & Failure Detection** and **Recovery Mechanism** Design in Navigation

(2) Why need to solve this problem?

- Agile Robot Navigation in **Uncertainties**
- **Guaranteeing Safety** and **Avoiding Collision** in Unknown environments
- Ensuring **Navigation Success**

(3) How is it different from prev.?

- Hard-Constrained Motion Planner with Proactive Failure Detection and Recovery Mechanism
- Model Agnostic Safety Monitoring Method
- Sim-to-Real Manner for Training Planner Failure Detector

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†**Disclaimer:** This summarization is for research and study purposes. It represents a personal interpretation and may contain inaccuracies. Feedback or corrections via email are highly appreciated.

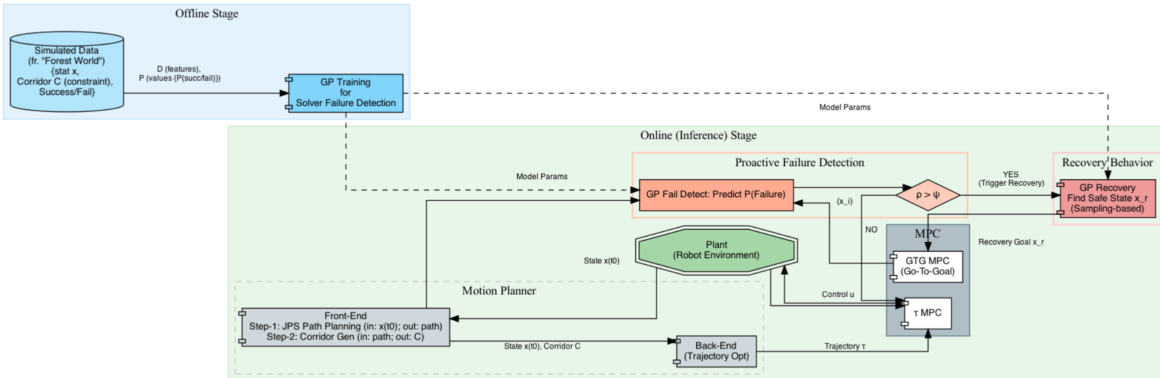

Figure 1: Summarized Block-Diagram of GP-based Robust Motion Planning

(4) Why is it better than prev.? (Advantages)

- Advantages among *Typical Hard-Constrained* Methods
  - Proactive Failure Detection enables **Adaptability to Disturbances**
  - **Recovery** Mechanism
  - Remain Rigor Hard Constrains while Incorporating Failure Processing Strategies (Safety Monitoring+Recovery)
- Advantages among *Soft-Constrained* Methods
  - Compliance with Hard Constraints ensures **Safety**
  - Generating Relatively **Higher Quality** and **Trackable** Trajectories
- Advantages among *Reachability Analysis* based Safety Monitoring Methods (Hamilton-Jacobi-Isaacs, ML based, etc.)
  - **Generalizability** and **Model Agnostic**
- Advantages among Other ML based Failure Detection (Safety Monitoring) Methods
  - **More robust**, **Non-Manual** Recovery Mechanism
  - **Computational Efficient** without Complex DL models
  - **Sim-to-Real** Manner: No Need of Exhaustive Real World Training Data

(5) What is the approach itself?

- **Gaussian Process** based **Proactive Failure Detection** in Motion Planning
- GP-based, sampling-based **Recovery** after Failure Detection
- **Sim-to-Real** Training Data for Training Failure Detector

(6) What are the applications of it? The approach can be demonstrated on several robotic platforms (eg. Boston Dynamics Spot quadruped, Jackal differential drive UGV) for safety-guaranteed and collision-free navigation in unknown environments.

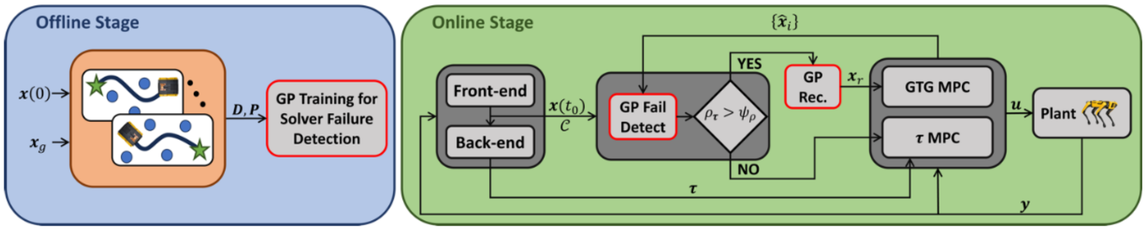

Figure 2: Original Block-Diagram in Paper

2 The Structure

Summarized Block-Diagram The summarized block-diagram see fig. 1[1].

Original in Paper The original block-diagram see fig. 2.

3 Open Questions

1. **Scalability and Representing Multi-Modal Distributions:** How does the proposed framework address the $O(N^3)$ computational complexity of Gaussian Processes (GPs) as the robot accumulates more environmental data? Furthermore, can a single Gaussian Process model effectively represent multi-modal failure risk distributions when the true risk landscape exhibits multiple distinct failure modes in topologically complex environments?

2. **Transition to Generative Normalizing Flows:** Could the current discriminative GP classifier for failure detection be augmented or replaced with a generative Normalizing Flow (NF) model that can capture multi-modal failure risk distributions? Specifically, would NF's latent space representation enable more efficient sampling of recovery candidate states compared to the current sampling approach?

References

N. Mohammad, J. Higgins, and N. Bezzo. A gp-based robust motion planning framework for agile autonomous robot navigation and recovery in unknown environments. In *2024 IEEE International Conference on Robotics and Automation (ICRA)*, pages 2418–2424. IEEE, 2024.

[1]This block-diagram is created through code-based drafting (Graphviz) for efficiency. While visual aesthetics may be limited, the overall structure remains clear.